Journal of Allbiosolution
(2022) Volume 4 Issue 1 Article ID: O0401016

Research article

Levels of heavy metals in uncontrolled agricultural farms compared to regulated organic farms

Jessica Briffa [1], Renald Blundell [2,3]*, Emmanuel Sinagra [4], Joseph Grech [5]

[1] Faculty of Medicine and Surgery, University of Malta, Msida, MSD 2080, Malta; jessica.mizzi.05@um.edu.mt
[2] Faculty of Medicine and Surgery, Centre for Molecular Medicine, Biomedical Science Building, University of Malta, Msida, MSD 2080, Malta;
[3] American University of Malta, Triq Dom Mintoff, Bormla, BML1013, Malta
[4] Faculty of Science, Chemistry and Pharmacology Building, University of Malta, Msida, MSD 2080, Malta;
[5] Faculty of Science, Chemistry and Pharmacology Building, University of Malta, Msida, MSD 2080, Malta
* Correspondence: renaldblundell@gmail.com

Abstract

Over the last few years, soil pollution was observed to have increased rapidly, mostly due to anthropogenic sources. Heavy metal pollution has been seen to enter the food chain which has caused great concern due to the adverse affect to the health of the public wellbeing. Anthropogenic activities include livestock and agriculture soil pollution; sewage, waste production and disposal; infrastructure, transportation and urbinanisation; industrial activity; fireworks; mining and warfare. Conventional farmers use a number of pesticides and fertilizers containing heavy metals which unfortunately remain in the soils and water table, and eventually end up in the crops and food chain. The aim of the study was to evaluate the difference between the amount of heavy metals present in conventional and organic farms and to observe if there is a significant difference between the two types of farming.

One hundred and one samples from around Malta and Gozo in 2019 were compared to two organic certified farms which confirmed through binomial testing that all ρ-values from the 101 samples compared with both controls, were less than the 0.05 level of significance stating that the district score differs significantly from the control value. Results showed that heavy metals where mostly found in the South-Eastern and South Harbour district. In the South-Eastern district, heavy metals found were vanadium at 64.25 mg/kg, manganese at 586.67 mg/kg, chromium at 45.93 mg/kg, selenium at 15.51 mg/kg and nickel at 21.90 mg/kg. In the Southern Harbour district, the heavy metals present were lead at 249.51 mg/kg, zinc at 251.33 mg/kg and copper at 95.15 mg/kg. The Northern district contained high concentrations of the heavy metals molybdenum at 12.09 mg/kg and arsenic at 7.93 mg/kg. In the Northern Harbour there is a high concentration of cadmium at 30.10 mg/kg. Cobalt was found highest in the Gozo and Comino district with a concentration of 15.8 mg/kg. The Finland standards were used to compare the results. Lead, cadmium, arsenic and zinc were to be found at high concentrations then the Finland standards, where the standards were at 60 mg/kg, 1 mg/kg, 5 mg/kg and 200 mg/kg respectively. Organic agricultural land proved to have lower concentrations of heavy metals than conventional agricultural land, which may be attributed to the lack of pesticide and chemical use.

Received: 21/06/2021, Accepted: 19/07/2021, Published:01/01/2022

Keywords

Agricultural land, organic farms, XRF, heavy metals, pollution, environment, soil

---

Introduction

Due to industrialization, natural resources have been exploitated to an exaggerated extent and have caused environmental pollution [1]. Heavy metals display toxicological properties due to their tendency of forming covalent bonds, therefore they can bind covalently with organic groups. Due to this important significance of this property, they form lipophilic ion and compounds, causing them to generate toxic effects when binding to the cellular macromolecules of nonmetallic elements. Due to the lipophilic property, the heavy metal is distributed within the biosphere differently from the simple ionic form of that same element. This toxic response also varies due to the lipophilic property. Lipophilic compounds include methylated forms of arsenic which are extremely toxic. Lead and mercury are examples of metalloids binding to nonmetallic elements such as sulphydryl groups of protein. There are four diverse ways for heavy metals to enter the human body; drinking contaminated water, ingestion of contaminated food, skin contact and inhalation from the atmosphere [2,3].

Heavy metal pollution is of concern due to the property that they cannot be broken down in the body and are nonbiodegradable. Detoxification of metal ions may occur in organisms by concealing the active element inside a protein or deposited in an insoluble form in the intracellular granules which will be excreted by the organism through the faeces or else will be stored long-term. When the heavy metals enter our body, they bioaccumulate in the body classifying them as dangerous. Biological and physiological problems are seen due to this bioaccumulation. Some of the metalloids are essential to life as they are required for a variety of functions, both physiological and biochemical [1,4–7]. Although essential, they are still toxic if present in large quantities. Heavy metals have been used broadly in agriculture, medicine, industry and other sectors and have thus been spread into the environment including our waters, soils and atmosphere [4–7].

Heavy metals have been reported to affect the cellular organelles together with the components of the cell. These include mitochondria, lysosomes, nuclei, enzymes and cell membrane. Metal ions have been observed to interact with nuclear proteins and DNA, leading to DNA damage and subsequently leading to cell cycle modulation, carcinogenesis or apoptosis [4].

Site-specific damage is caused when heavy metals interact with both the DNA and he nuclear proteins, causing ‘indirect’ or ‘direct’ damage. ‘Indirect” damage is caused by the production of nitrogen species and reactive oxygen which comprise of superoxide and hydroxyl radicals, nitric oxide, hydrogen peroxide and other endogenous oxidants. ‘Direct’ damage causes conformation changes to the biomolecules due to the heavy metal. Heavy metals have been noted to activate signaling pathways. Free radicals are formed from heavy metals causing DNA damage, lipid peroxidation and the alteration of the sulphydryl homestasis. Metal-mediated calcium homeostasis has also been noted to be altered due to membrane damage causing a number if calcium dependent systems to become activated including endonucleases. Iron, nickel, chromium, copper and cadmium have been investigated due to the formation of free radicals, where cadmium, nickel and chromium are observed to have carcinogenic properties [8,9].

Sources of anthropogenic pollution vary from livestock and agriculture; industrialization; infrastructure, including transport and urbanization; sewage, waste production and disposal; mining and also warfare. A lot of farmers over-fertilize, using pesticides and herbicides, untreated wastewater or inferior quality water for irrigation, therefore increasing the risk of heavy metal pollution in the

Received: 21/06/2021, Accepted: 19/07/2021, Published:01/01/2022

soils [10,11]. Fertilizers are added in excess by most farmers leading to a surplus of nitrogen and phosphorous in the soil [10]. It can also produce a heavy metal build-up and an increase in the soil's salinity [12]. Mismanagement of fertilizer was seen in Northeast China where diffuse pollution was observed contributing to environmental damage causing national and international alarm [13]. The increase in nitrogen may be lost to the atmosphere due to green gas emissions, while excess phosphorous may cause the eutrophication of water sources found nearby. Heavy metals such as cadmium, chromium, cobalt, nickel and lead, together with natural radionuclides are also added to the soil by the use of excess fertilizer depending on which country the fertilizer has been compounded [11,14–16].

Recently studies have shown that livestock waste is seen as a cause of soil pollution if it is not removed correctly. Animal manure is an essential source of nutrients for plants and soil, and its use in agriculture has increased. Though evidence has shown that animal manure may increase soil pollution due to the presence of heavy metals, pathogens and veterinary antibiotic residues, which may produce an increase in antimicrobial-resistant bacteria in soils [10]. The animal urine and faeces contain microorganisms which can be pathogenic or nonpathogenic to humans and animals. The microorganism found in the soil will be negatively affected due to the livestock waste added. Veterinary medicines tend to be excreted by the livestock which do not degrade quickly due to its lipolytic properties, therefore causing the manure to retain the medial substance which will eventually end up in the soil [17–19].

Pesticides are commonly used on crops to control or eradicate the pest or weeds which interfere with crop growth. Furthermore pesticides aid in the reduction of disease which affects the growth of the harvest and reduces the harvest by 40% [20]. A negative impact on the health of both humans and the environment has been seen due to the pesticides being used [10,21]. A decline in the number of microorganisms and fungi that can be found naturally in soil have been decreasing due to the use of pesticides, leading to the degradation of soil. Pesticides can also be leached off ending up in the drinking water supply proving to be hazardous [11,22].

Heavy metal contamination in the soil, food scares and disease has led to a boom in the buying of organic foods from organic farming. A study was done to compare the concentrations of heavy metals in soil of two different types of agricultural soil, conventional and organic, and note if there are any significant differences between these types of farming.

2. Method and Materials

One hundred and one samples from conventional farmland were collected from around Malta and Gozo through stratified sampling. Samples were taken when possible from each locality to ensure representation. Samples were collected between Februaury 2019 and August 2019. Twent two localities could not be sued as either no agricultural soil was present or permission was not granted to test their soil.

Each sample contained soil from the four corners of the land together with a central one. Tope vegetation and debirs was removed before digging up the soil, and holes were dug up to 12 inches deep. The uprooted soils were thoroughly mixed to obtain a homogenous sample. Soil samples were placed in sterilized plastic containers, and all equipment used like hand trowel was made of plastic not to contaminate the sample with metals. The hand trowel was cleaned after every use so as not to cross-contaminate.

Received: 21/06/2021, Accepted: 19/07/2021, Published:01/01/2022

At the laboratory, these samples were put in plastic trays, mixed and air-dried at ambient temperature. The samples were then sieved using a 45 micron mesh to remove any debris and stones and homogenize the soil sample. The samples were then placed in watch glasses and oven-dried for 24 hours at ~110°C. The standard protocol was followed with proper care so that the sample cross-contamination and external contamination, would not occur. The soil samples were sieved for a second time to make it finer and were then stored in sterilized plastic containers containing desiccators [23].

Heavy metal concentrations were determined using the S2 Ranger Energy Dispersive X-Ray Fluorescence (EDXRF) spectrometer situated at the laboratory of the Department of Chemistry, at the University of Malta. Before each batch of the sample was placed in the instrument, the XRF spectrometer was calibrated. A copper disc calibration was analyzed first, followed by a quality check. The sample cup method was used to analyze the soil. The sample cup was assembled using a 3.6 µm SpectroMembrane® Mylar® Thin-film. The film was examined to see if it is was taut enough to give accurate readings. Soil was poured gently into the cup, tapped and compressed to make the soil more compact and be absent of air pockets. The lid was fastened onto the cup and labelled accordingly. The lid of the XRF was opened and each cup was placed on a ring, and then the lid was closed. Calcium carbonate was chosen as the matrix and the elements were expressed as oxides. Each round of analyzes consisted of fourteen samples, where each sample took an average of twelve minutes to be analysed. The data produced by the instrument was semi-quantitative, and the elemental concentrations were then established by using a fitting method supplied by Bruker. Values for each test was to have a fitting parameter of R/R_0 of less then 30, as the lower the fitting parameter, the better the fit is. The instrument's software used to read the data, Spectra EDX Launcher [23].

Each sample was tested three times to ensure that the spectrometer was providing reliable results. The statistical analysis was achieved using the IBM® SPSS® Statistics 26 software package. Tests used to ensure the accuracy of the results were the Shapiro-Wilk test, the One-Way ANOVA, together with the Kruskal-Wallis test. These would determine if the elemental score distribution in the samples are significant or not.

The Shapiro-Wilk test was applied to determine if the element score distribution is normal or skewed. The null hypothesis states that the score distribution is normal and in the not rejected if the ρ-value exceed the level of significance. The alternative hypothesis states that the score distribution if skewed and is not rejected if the ρ-value is less than the 0.05 level ofsignificance. If the score distribution is skewed, both the One-Way ANOVA and the Kruskal-Wallis test would be applied to compare the elemental scores between the five samples. When the there is a normal elemental score distribution the parametric test One-Way ANOVA is used, whilst when the score distribution is skewed, the non-parametric test Kruskal-Wallis test is used. The null hypothesis for both tests states that the mean lement scores differ marginally between the samples and is rejected if the ρ-value exceeds the level of significance. When the elemental scores varies significantly between the samples, the alternative hypotheses states that the results is not rejected if the ρ-value is less than the 0.05 criteria.

After collecting and analyzing the one hundred and one samples, two more samples were collected from organic farms. The difference in these samples is that no pesticides or contaminated water or manure is used. The estates hold an Organic Farming Council Regulation (EC) No 834/2007 accoridng to the EU Agriculture. These results were then compated to the one hundred and one samples where a statistical analysis was determined using the software package IBM® SPSS® Statistics 26 to observe if there is a significant difference between conventional and organi farming. The data were compared by using the binomial test to calculate the ρ-value and its level of significance. Four possible outcomes could result from these comparisons.

Received: 21/06/2021, Accepted: 19/07/2021, Published:01/01/2022

1) In both binomial tests, the null hypothesis is not rejected meaning that the sample mean of the 101 observations varies marginally from the control one value and the control two value.
2) In both binomial tests, the alternative hypothesis is not rejected meaning that the sample mean of the 101 observations varies significantly from the control one value and the control two value.
3) The null hypothesis is not rejected in the first binomial test, and the alternative hypothesis is not rejected in the second Binomial test meaning that the sample mean of the 101 observations vary marginally from the control one value, but vary significantly from the control two value.
4) The alternative hypothesis is not rejected in the first binomial test, and the null hypothesis is not rejected in the second Binomial test meaning that the sample means of the 101 observations varies significantly from the control one value, but varies marginally from the control two value.

Results and Discussion

All the one hundred and one samples were tested three times each so as to see if there was consistency in the data given. The Shapiro-Wilk test was used which showed that the ρ-values of some results exceeded the 0.05 level of significance, therefore indicating that the element score distribution is normal. For the other results the ρ-value was less than the 0.05 criteria therefore indicating that the element score distribution was scored and thus did not satisfy the normality assumption (table 1). Due to this, both parametric (One-Way ANOVA) and non-parametric (Kruskal-Wallis) tests had to be used to analyze the data. Repetitive sample was done to compare the data and observe if the readings differ from one run to another. The overall results showed that most of the samples had a level of significance of more than 0.05, therefore there was only a minimal difference between the same sample testing. Only a small amount of the samples had less than 0.05 level of significance, therefore resulting that the sample results varied marginally when repeated (table 2). Overall, the results showed that there was no difference in the results and the data presented was accurate. Chromium peaks were seen to be always present in the graphs though on calculating the R/R_0 through the software programme to get a good fit, the data occasionally did not present chromium as one of the metals present, even though present. This also occurred to nickel and vanadium.

After confirming the results, the mean score of these one hundred and one samples were compared to the two samples from the organic farming (control 1 and control 2) by using the binomial test. The results showed that all ρ-values, with both organic soil samples, were less than the 0.05 level of significance stating that the conventional soil values differ significantly from the organic soil values (table 3 and table 4). The null hypothesis of both binomial tests is not rejected meaning that the sample mean of the one hundred and one values varies marginally from the control 1 value and the control 2 value.

H1 was not rejected in both binomial tests as the sample mean of the one hundred and one valuesnvaries significantly from both the control values. As seen in table 4, it proved that most elements were statistically different in both organic farmland, compared to the other one hundred

Table 1: Shapiro-Wilk Test

	Statistics	Shapiro-Wilk df	ρ-value
Aluminium	0.838	15.000	0.012
Arsenic	0.713	15.000	0.000
Barium	0.719	15.000	0.000

Received: 21/06/2021, Accepted: 19/07/2021, Published:01/01/2022

Bismuth	0.895	15.000	0.080
Bromine	0.896	15.000	0.082
Calcium	0.916	15.000	0.168
Cadmium	0.828	15.000	0.008
Cerium	0.000	15.000	0.000
Chlorine	0.927	15.000	0.242
Cobalt	0.284	15.000	0.000
Chromium	0.000	15.000	0.000
Caesium	0.524	15.000	0.000
Copper	0.967	15.000	0.811
Iron	0.841	15.000	0.013
Gallium	0.827	15.000	0.008
Germanium	0.665	15.000	0.000
Iodine	0.819	15.000	0.007
Potassium	0.952	15.000	0.563
Lanthanum	0.284	15.000	0.000
Magnesium	0.838	15.000	0.012
Manganese	0.907	15.000	0.123
Molybdenum	0.899	15.000	0.091
Sodium	0.828	15.000	0.009
Niobium	0.909	15.000	0.132
Neodymium	0.000	15.000	0.000
Nickel	0.729	15.000	0.001
Phosphorus	0.965	15.000	0.779
Lead	0.919	15.000	0.188
Praseodymium	0.000	15.000	0.000
Rubidium	0.914	15.000	0.155
Antimony	0.816	15.000	0.006
Selenium	0.833	15.000	0.010
Silicon	0.868	15.000	0.031
Tin	0.839	15.000	0.012
Sulphur	0.890	15.000	0.068
Strontium	0.603	15.000	0.000
Tantalum	0.000	15.000	0.000
Tellurium	0.827	15.000	0.008
Titanium	0.787	15.000	0.002
Vanadium	0.604	15.000	0.000
Tungsten	0.000	15.000	0.000
Yttrium	0.911	15.000	0.139
Zinc	0.865	15.000	0.028
Zirconium	0.930	15.000	0.271

Table 2: One-Way ANOVA

		Sample Size	Mean	Std. Deviation	p-value
Aluminium	A	3.000	26921.183	319.017	0.012
	B	3.000	28844.125	1455.197	
	C	3.000	21734.533	2320.466	
	D	3.000	13337.100	521.251	
	E	3.000	28791.200	1894.242	
Arsenic	A	3.000	7.574	13.118	0.528
	B	3.000	12.623	11.344	
	C	3.000	6.564	11.369	
	D	3.000	0.000	0.000	

Received: 21/06/2021, Accepted: 19/07/2021, Published:01/01/2022

	E	3.000	14.643	13.234	
Barium	A	3.000	208.388	360.938	0.494
	B	3.000	440.361	402.439	
	C	3.000	221.226	383.174	
	D	3.000	0.000	0.000	
	E	3.000	392.593	387.905	
Bismuth	A	3.000	32.890	18.148	0.141
	B	3.000	37.674	16.805	
	C	3.000	33.787	16.278	
	D	3.000	14.950	2.740	
	E	3.000	43.056	17.509	
Bromine	A	3.000	36.000	13.856	0.083
	B	3.000	48.333	13.577	
	C	3.000	41.667	14.434	
	D	3.000	29.667	1.528	
	E	3.000	68.333	13.013	
Calcium	A	3.000	226707.887	2576.655	0.035
	B	3.000	226440.700	8670.197	
	C	3.000	243273.460	15970.970	
	D	3.000	271328.060	1746.960	
	E	3.000	208672.787	20104.354	
Cadmium	A	3.000	24.010	21.051	0.112
	B	3.000	37.478	23.510	
	C	3.000	37.478	24.850	
	D	3.000	3.806	0.507	
	E	3.000	43.334	15.623	
Cerium	A	3.000	0.000	0.000	1.000
	B	3.000	0.000	0.000	
	C	3.000	0.000	0.000	
	D	3.000	0.000	0.000	
	E	3.000	0.000	0.000	
Chlorine	A	3.000	2350.000	374.032	0.025
	B	3.000	2496.667	338.575	
	C	3.000	2376.667	304.357	
	D	3.000	1853.333	92.916	
	E	3.000	3290.000	470.319	
Cobalt	A	3.000	0.000	0.000	0.406
	B	3.000	0.000	0.000	
	C	3.000	3.932	6.811	
	D	3.000	0.000	0.000	
	E	3.000	0.000	0.000	

		Sample Size	Mean	Std. Deviation	p-value
Chromium	A	3.000	0.000	0.000	1.000
	B	3.000	0.000	0.000	
	C	3.000	0.000	0.000	
	D	3.000	0.000	0.000	
	E	3.000	0.000	0.000	
Caesium	A	3.000	0.000	0.000	0.190
	B	3.000	77.929	74.410	
	C	3.000	0.000	0.000	
	D	3.000	0.000	0.000	
	E	3.000	46.588	80.692	

Received: 21/06/2021, Accepted: 19/07/2021, Published:01/01/2022

Copper	A	3.000	96.928	16.782	0.133
	B	3.000	97.461	16.833	
	C	3.000	101.988	17.726	
	D	3.000	59.648	12.840	
	E	3.000	101.722	26.929	
Iron	A	3.000	23640.734	209.829	0.009
	B	3.000	25482.566	345.021	
	C	3.000	17089.406	213.679	
	D	3.000	11400.709	252.183	
	E	3.000	26764.855	672.084	
Gallium	A	3.000	13.391	12.690	0.190
	B	3.000	15.127	14.239	
	C	3.000	7.439	12.885	
	D	3.000	0.000	0.000	
	E	3.000	18.846	13.021	
Germanium	A	3.000	1.851	1.747	0.187
	B	3.000	7.405	7.292	
	C	3.000	0.000	0.000	
	D	3.000	0.000	0.000	
	E	3.000	3.240	3.947	
Iodine	A	3.000	41.000	57.297	0.918
	B	3.000	72.000	65.483	
	C	3.000	71.333	51.394	
	D	3.000	14.333	5.132	
	E	3.000	64.667	62.003	
Potassium	A	3.000	9491.382	1631.690	0.018
	B	3.000	11400.727	2129.999	
	C	3.000	7867.055	743.730	
	D	3.000	4983.667	481.081	
	E	3.000	11566.757	2786.057	
Lanthanum	A	3.000	0.000	0.000	0.406
	B	3.000	0.000	0.000	
	C	3.000	82.142	142.273	
	D	3.000	0.000	0.000	
	E	3.000	0.000	0.000	
Magnesium	A	3.000	14372.453	1055.464	0.305
	B	3.000	14493.061	1268.297	
	C	3.000	15658.939	4228.310	
	D	3.000	15819.749	857.788	
	E	3.000	18593.733	4342.028	

		Sample Size	Mean	Std. Deviation	p-value
Manganese	A	3.000	447.896	82.636	0.025
	B	3.000	527.149	71.755	
	C	3.000	343.602	88.622	
	D	3.000	143.017	16.652	
	E	3.000	546.252	62.752	
Molybdenum	A	3.000	12.664	8.081	0.186
	B	3.000	15.108	7.668	
	C	3.000	13.331	9.236	
	D	3.000	2.666	0.667	
	E	3.000	16.219	5.823	

Received: 21/06/2021, Accepted: 19/07/2021, Published:01/01/2022

Sodium	A	3.000	98667.380	3709.300	0.081
	B	3.000	90754.207	2384.746	
	C	3.000	117461.167	17825.235	
	D	3.000	115977.447	3659.508	
	E	3.000	117708.453	40186.176	
Niobium	A	3.000	21.437	11.536	0.125
	B	3.000	27.030	11.911	
	C	3.000	23.534	13.121	
	D	3.000	5.592	0.699	
	E	3.000	27.962	9.088	
Neodymium	A	3.000	0.000	0.000	1.000
	B	3.000	0.000	0.000	
	C	3.000	0.000	0.000	
	D	3.000	0.000	0.000	
	E	3.000	0.000	0.000	
Nickel	A	3.000	11.263	19.508	0.380
	B	3.000	18.859	20.142	
	C	3.000	4.453	7.713	
	D	3.000	0.000	0.000	
	E	3.000	22.526	22.021	
Phosphorus	A	3.000	4109.622	402.651	0.131
	B	3.000	4236.183	572.382	
	C	3.000	3069.487	309.284	
	D	3.000	3895.776	64.338	
	E	3.000	4067.434	631.408	
Lead	A	3.000	131.512	19.078	0.031
	B	3.000	94.998	17.025	
	C	3.000	76.122	16.736	
	D	3.000	153.482	1.418	
	E	3.000	101.496	19.852	
Praseodymium	A	3.000	0.000	0.000	1.000
	B	3.000	0.000	0.000	
	C	3.000	0.000	0.000	
	D	3.000	0.000	0.000	
	E	3.000	0.000	0.000	
Rubidium	A	3.000	69.800	10.072	0.022
	B	3.000	78.334	9.777	
	C	3.000	53.341	9.777	
	D	3.000	32.004	1.584	
	E	3.000	80.163	9.159	
Antimony	A	3.000	64.322	61.663	0.735
	B	3.000	97.179	61.535	
	C	3.000	91.332	53.111	
	D	3.000	31.743	5.081	
	E	3.000	48.172	83.436	
		Sample Size	**Mean**	**Std. Deviation**	**p-value**
Selenium	A	3.000	14.707	10.682	0.154
	B	3.000	18.977	11.570	
	C	3.000	17.553	9.818	
	D	3.000	5.693	0.712	
	E	3.000	20.400	10.706	
Silicon	A	3.000	100871.000	1471.273	0.026
	B	3.000	100658.640	7724.184	
	C	3.000	72457.232	13475.610	

Received: 21/06/2021, Accepted: 19/07/2021, Published:01/01/2022

	D	3.000	48481.788	2111.995	
	E	3.000	98959.760	12293.058	
Tin	A	3.000	129.442	31.451	0.168
	B	3.000	153.860	34.893	
	C	3.000	145.721	24.671	
	D	3.000	107.650	0.455	
	E	3.000	147.034	30.723	
Sulphur	A	3.000	1764.870	281.216	0.045
	B	3.000	2035.875	378.728	
	C	3.000	1788.900	96.148	
	D	3.000	1495.200	128.306	
	E	3.000	2690.025	488.911	
Strontium	A	3.000	341.900	4.808	0.014
	B	3.000	346.410	1.292	
	C	3.000	337.672	3.996	
	D	3.000	503.972	3.686	
	E	3.000	355.993	11.186	
Tantalum	A	3.000	0.000	0.000	1.000
	B	3.000	0.000	0.000	
	C	3.000	0.000	0.000	
	D	3.000	0.000	0.000	
	E	3.000	0.000	0.000	
Tellurium	A	3.000	53.567	51.739	0.925
	B	3.000	74.887	54.661	
	C	3.000	71.422	46.648	
	D	3.000	21.320	2.808	
	E	3.000	63.960	54.325	
Titanium	A	3.000	2697.030	27.465	0.038
	B	3.000	2802.913	30.166	
	C	3.000	2269.501	677.290	
	D	3.000	739.186	136.714	
	E	3.000	2978.720	213.082	
Vanadium	A	3.000	0.000	0.000	0.277
	B	3.000	22.594	37.210	
	C	3.000	24.461	42.367	
	D	3.000	0.000	0.000	
	E	3.000	38.838	46.764	
Tungsten	A	3.000	0.000	0.000	1.000
	B	3.000	0.000	0.000	
	C	3.000	0.000	0.000	
	D	3.000	0.000	0.000	
	E	3.000	0.000	0.000	

		Sample Size	Mean	Std. Deviation	p-value
Yttrium	A	3.000	43.309	11.759	0.080
	B	3.000	51.184	11.600	
	C	3.000	40.947	13.662	
	D	3.000	23.886	0.909	
	E	3.000	53.283	7.927	
Zinc	A	3.000	177.551	3.502	0.016
	B	3.000	180.765	8.946	
	C	3.000	104.442	7.004	
	D	3.000	131.758	4.473	

Received: 21/06/2021, Accepted: 19/07/2021, Published:01/01/2022

	E	3.000	189.870	8.362	
Zirconium	A	3.000	342.521	106.661	0.084
	B	3.000	397.058	109.130	
	C	3.000	325.494	119.052	
	D	3.000	84.150	8.770	
	E	3.000	417.047	110.321	

Table 3: Binomial analysis between the mean values of the 101 samples and control 1

Element	Symbol	Number of Samples	Mean	Mean St. Dev	Control 1	ρ-value
Aluminium	**Al**	101	35447.17	10740.71	20217.35	0
Arsenic	**As**	101	5.63	8.959	1.2	0.005
Bismuth	**Bi**	101	29.41	14.578	7.005492	0
Cadmium	**Cd**	101	24.01	20.523	16	0.111
Cerium	**Ce**	101	0.79	7.938	0	0
Cobalt	**Co**	101	6.98	19.877	5.62	0
Chromium	**Cr**	101	23.98	51.071	0	0
Caesium	**Cs**	101	4.72	23.806	0	0
Copper	**Cu**	101	65.17	63.605	65.31	0.691
Iron	**Fe**	101	26994.54	10158.57	20238	0
Gallium	**Ga**	101	12.57	14.16	4.664441	0.32
Germanium	**Ge**	101	2.45	5.05	0	0
Lanthanum	**La**	101	29.11	105.111	0	0
Manganese	**Mn**	101	487.89	248.112	389.21	0.017
Molybdenum	**Mo**	101	9.16	7.661	1.093126	0
Niobium	**Nb**	101	19.84	12.822	5.997763	0
Neodymium	**Nd**	101	6.51	37.041	0	0
Nickel	**Ni**	101	14.4	20.73	0	0.111
Lead	**Pb**	101	131.01	105.713	78.9072	0.017
Praseodymium	**Pr**	101	0	0	0	0
Rubidium	**Rb**	101	77.51	31.142	74.06721	0.32
Antimony	**Sb**	101	47.1	51.059	36.2709	0.001
Selenium	**Se**	101	12.63	8.832	11.385	0.009
Tin	**Sn**	101	131.4	28.788	127.5445	0.32
Strontium	**Sr**	101	309.42	105.595	213.0887	0
Tantalum	**Ta**	101	0	0	0	0
Tellurium	**Te**	101	44.95	41.347	24.00099	0.073
Titanium	**Ti**	101	2797.16	1074.09	2016.27	0
Vanadium	**V**	101	35.75	91.9	0	0
Tungsten	**W**	101	0.4	2.709	0	0
Yttrium	**Y**	101	33.78	14.781	9.260294	0
Zirconium	**Zr**	101	306.16	162.415	254.18	0.551

Table 4: Binomial analysis between the mean values of the 101 samples and control 2

Element	Symbol	Number of Samples	Mean	Mean St. Dev	Control 2	ρ-value

Received: 21/06/2021, Accepted: 19/07/2021, Published:01/01/2022

Aluminium	**Al**	101	35447.17	10740.71	23014.35	0
Arsenic	**As**	101	5.63	8.959	0.8	0.005
Bismuth	**Bi**	101	29.41	14.578	10	0
Cadmium	**Cd**	101	24.01	20.523	10	0.003
Cerium	**Ce**	101	0.79	7.938	0	0
Cobalt	**Co**	101	6.98	19.877	5	0
Chromium	**Cr**	101	23.98	51.071	0	0
Caesium	**Cs**	101	4.72	23.806	0	0
Copper	**Cu**	101	65.17	53.605	48.73046	0.32
Iron	**Fe**	101	26994.54	10158.57	23000	0.005
Gallium	**Ga**	101	12.57	14.16	5.01	0.32
Germanium	**Ge**	101	2.45	5.05	0	0
Potassium	**K**	101	10368.57	3294.815	12038	0
Lanthanum	**La**	101	29.11	105.111	0	0
Manganese	**Mn**	101	487.89	248.112	352.64	0
Molybdenum	**Mo**	101	9.16	7.661	1.53	0
Niobium	**Nb**	101	19.84	12.8222	5.62	0
Neodymium	**Nd**	101	6.51	37.041	0	0
Nickel	**Ni**	101	14.4	20.73	0	0.111
Lead	**Pb**	101	131.01	105.713	76.1224	0.005
Praseodymium	**Pr**	101	0	0	0	0
Rubidium	**Rb**	101	77.51	31.142	75.0348	0.232
Antimony	**Sb**	101	47.1	51.059	30.246	0.551
Selenium	**Se**	101	12.63	8.832	10.24	0.046
Tin	**Sn**	101	131.4	28.788	125.876	0.691
Strontium	**Sr**	101	309.42	105.595	146.287	0
Tantalum	**Ta**	101	0	0	0	0
Tellurium	**Te**	101	44.95	41.347	0	0
Titanium	**Ti**	101	2797.16	1074.09	1958.6	0
Vanadium	**V**	101	35.75	91.9	0	0
Tungsten	**W**	101	0.4	2.709	0	0
Yttrium	**Y**	101	33.78	14.781	15.021	0
Zirconium	**Zr**	101	306.16	162.415	201.65	0

Table 1: Threshold and guideline values for harmful substances in soil according to the Ministry of Environment, Finland [24].

Substance (symbol	**Natural Concentration[1] mg/kg**	**Threshold Value mg/kg**	**Lower guideline value mg/kg**	**Higher guideline value mg/kg**
Antinomy (Sb) (p)	0.02 (0.01-0.2)	2	10 (t)	50 (e)
Arsenic (As) (p)	1 (0.1-25)	5	50 (e)	100 (e)
Mercury (Hg)	0.005 (<0.005-0.05)	0.5	2 (e)	5 (e)
Cadmium (Cd)	0.02 (0.01-0.15)	1	10 (e)	20 (e)
Cobalt (Co) (p)	8 (1-30)	20	100 (e)	250 (e)
Chromium (Cr)	31(6-170)	100	200 (e)	300 (e)

Received: 21/06/2021, Accepted: 19/07/2021, Published:01/01/2022

Copper (Cu)	22 (5-110)	100	150 (e)	200 (e)
Lead (Pb)	5 (0.1-5)	60	200 (t)	750 (e)
Nickel (Ni)	17 (3-100)	50	100 (e)	150 (e)
Zinc (Zn)	31 (8-110)	200	250 (e)	400 (e)
Vanadium (V)	38 (10-115)	100	150 (e)	250 (e)

[1] The median and range of the natural concentration of fines in moraine when defined by extraction with aqua regia, except pyrolytically defined mercury. It must be taken into account in site-specific analyses that especially in clay soils the natural concentrations may be clearly higher than those measured from moraine.
(e) – Ecological Risk; (t) – Health Risks

and one farms and show that they contain fewer concentrations of heavy metals. In control 1, Cu is not significantly different since the ρ-value exceeds the 0.05 level of significance. This was proved to be the case as the farmer was using copper as a fungicide thus increasing the copper concentration in the soil. On the other hand, in control 2, chlorine was not statistically significant as the ρ-value exceeds the 0.05 level of significance. The reason is that the farmland is very close to the seaside, where strong winds bring in sea droplets increasing the chlorine in the soil.

The Ministry of Environment of Finland has stated the threshold and guideline values for harmful substances in the soil (table 5). These guidelines were used as they are the only ones found when researching. In the conventional farms, on average, As and Pb were higher than the stipulated threshold guideline, whilst Cd was higher than the higher guideline value. In organic farming, none of the elements went higher than the threshold value. In the organic farms only Cd was found to be slightly higher than the guidelines.

In control 1, out of the forthy three elements tested, eight elements were found to have a ρ-value which exceeds the 0.05 level of significance (Cd, Cu, Ga, Ni, Rb, Sn, Te, and Zr), whilst in control 2, six elements were found to have a ρ-value which exceeds the 0.05 level of significance (Cu, Ga, Ni, Rb, Sb and Sn). The organic farms that were tested did not use any pesticides, using natural methods of preventing or killing pests such as soap and water mixtures, and they used animal fertilizer coming from organically fed cows. These reduce the chemicals in the food production industry and thus reduces medical complications and thus decreases the burden of medication and hospitalization. Biodiversity is increased when organic farming is practised and this likewise helps the natural process of eliminating pests [24–26].

Conclusion

Lately heavy metal pollution has attracted a lot of attention due to their toxicity, being biodegradable and will remain in the soil indefinitely. Heavy metal bioaccumulation in the soil can be of detriment to humans, animals and plants, since the heavy metals are first absorbed by the plant and will then end up in the food chain causing adverse effects including death [11].

Organic farming describes the production system that supports the health of ecosystems, soils and people. It counts on the ecological processes, cycles and biodiversity according to the local conditions, instead of the use of inputs like pesticides which have adverse effects. Organic farming unites tradition, science and innovation to help the communal environment and encourage fair relationships and a superior quality of life for all those concerned [27].

The study proved that organic agricultural land had fewer heavy metals in the soil and heavy metals with a less concentration than conventional agricultural worked land. This signifies that the crops grown in the soil of organic farms will contain lesser amounts too. Conventional farms still use pesticides and fertilizers containing heavy emtals. Phytoremediation and intercropping are popular inexpensive methods that are environmentally friendly and are being used to accumulate heavy

Received: 21/06/2021, Accepted: 19/07/2021, Published:01/01/2022

metals in the soil to lessen the concentration of heavy metals present in the soil. The decrease in the heavy metal bioaccumulation in the soil will help to reduce the concentration absorbed by the plants, and thus less will enter the food chain, reducing the adverse effects caused.

Funding

The research work disclosed in this publication is partially funded by the Endeavour Scholarship Scheme (Malta). Scholarships are part-financed by the European Union – European Social Fund (ESF) – Operational Programme II – Cohesian Policy 2014-20 "Investing in human capital to create more opportunities and promote to well-being of society".

References

[1] P.K. Gautam, R.K. Gautam, M.C. Chattopadhyaya, S. Banerjee, M.C. Chattopadhyaya, J.D. Pandey, Heavy metals in the environment: Fate, transport, toxicity and remediation technologies Thermodynamic profiling of pollutants View project Materials for Solid oxide fuel cells View project Heavy Metals In The Environment: Fate, Transport, Toxicity And Rem, 2016. https://www.researchgate.net/publication/314465070 (accessed November 1, 2019).

[2] V. Masindi, K.L. Muedi, Environmental Contamination by Heavy Metals, in: Heavy Met., InTech, 2018. https://doi.org/10.5772/intechopen.76082.

[3] D.B.P. C.H. Walker, R.M. Sibly, S.P. Hopkin, Principles Of Ecotoxicology, 4th Editio, CRC Press, 2012.

[4] P.B. Tchounwou, C.G. Yedjou, A.K. Patlolla, D.J. Sutton, Heavy Metal Toxicity and the Environment, in: EXS, 2012: pp. 133–164. https://doi.org/10.1007/978-3-7643-8340-4_6.

[5] Lenntech BV, Heavy Metals, (n.d.). https://www.lenntech.com/processes/heavy/heavy-metals/heavy-metals.htm (accessed June 29, 2018).

[6] J.H. Duffus, "Heavy metals" a meaningless term?, Pure Appl. Chem. 74 (2002) 793–807. https://doi.org/10.1351/pac200274050793.

[7] L.K. Wang, Heavy metals in the environment, CRC Press, 2009.

[8] M. Valko, H. Morris, M. Cronin, Metals, Toxicity and Oxidative Stress, Curr. Med. Chem. 12 (2005) 1161–1208. https://doi.org/10.2174/0929867053764635.

[9] J. Briffa, E. Sinagra, R. Blundell, Heavy metal pollution in the environment and their toxicological effects on humans, Heliyon. 6 (2020) e04691. https://doi.org/10.1016/j.heliyon.2020.e04691.

[10] N. Rodriguez-Eugenio, M. McLaughlin, D. Pennock, Soil Pollution: A Hidden Reality, Food and Agriculture Organization of the United Nations, Rome, 2018.

[11] J. Briffa, E. Sinagra, R. Blundell, Heavy metal pollution in the environment and their toxicological effects on humans, Heliyon. 6 (2020). https://doi.org/10.1016/j.heliyon.2020.e04691.

[12] D.R. Kanter, Nitrogen pollution: a key building block for addressing climate change, Clim. Change. 147 (2018) 11–21. https://doi.org/10.1007/s10584-017-2126-6.

[13] X.T. Ju, C.L. Kou, P. Christie, Z.X. Dou, F.S. Zhang, Changes in the soil environment from excessive application of fertilizers and manures to two contrasting intensive cropping systems

Received: 21/06/2021, Accepted: 19/07/2021, Published:01/01/2022

on the North China Plain, Environ. Pollut. 145 (2007) 497–506. https://doi.org/10.1016/j.envpol.2006.04.017.

[14] W.M. Stewart, D.W. Dibb, A.E. Johnston, T.J. Smyth, The contribution of commercial fertilizer nutrients to food production, Agron. J. 97 (2005) 1–6. https://doi.org/10.2134/agronj2005.0001.

[15] S. Savci, An Agricultural Pollutant: Chemical Fertilizer, Int. J. Environ. Sci. Dev. (2012) 73–80. https://doi.org/10.7763/ijesd.2012.v3.191.

[16] A.S. Modaihsh, M.S. Al-Swailem, M.O. Mahjoub, View of Heavy Metals Content of Commercial Inorganic Fertilizers Used in the Kingdom of Saudi Arabia, Agric. Mar. Sci. 9 (2004) 21–25. https://journals.squ.edu.om/index.php/jams/article/view/621/620 (accessed August 16, 2021).

[17] B. Halling-Sørensen, S. Nors Nielsen, P.F. Lanzky, F. Ingerslev, H.C. Holten Lützhøft, S.E. Jørgensen, Occurrence, fate and effects of pharmaceutical substances in the environment- A review, Chemosphere. 36 (1998) 357–393. https://doi.org/10.1016/S0045-6535(97)00354-8.

[18] H. Zhang, Y. Luo, L. Wu, Y. Huang, P. Christie, Residues and potential ecological risks of veterinary antibiotics in manures and composts associated with protected vegetable farming, Environ. Sci. Pollut. Res. 22 (2015) 5908–5918. https://doi.org/10.1007/s11356-014-3731-9.

[19] C.E. Manyi-Loh, S.N. Mamphweli, E.L. Meyer, G. Makaka, M. Simon, A.I. Okoh, An overview of the control of bacterial pathogens in cattle manure, Int. J. Environ. Res. Public Health. 13 (2016). https://doi.org/10.3390/ijerph13090843.

[20] I. Mahmood, S.R. Imadi, K. Shazadi, A. Gul, K.R. Hakeem, Effects of pesticides on environment, in: Plant, Soil Microbes Vol. 1 Implic. Crop Sci., Springer International Publishing, Cham, 2016: pp. 253–269. https://doi.org/10.1007/978-3-319-27455-3_13.

[21] J. Popp, K. Pető, J. Nagy, Pesticide productivity and food security. A review, Agron. Sustain. Dev. 33 (2013) 243–255. https://doi.org/10.1007/s13593-012-0105-x.

[22] P.H. Nicholls, Factors influencing entry of pesticides into soil water, Pestic. Sci. 22 (1988) 123–137. https://doi.org/10.1002/ps.2780220204.

[23] E. Agrafioti, D. Kalderis, E. Diamadopoulos, Arsenic and chromium removal from water using biochars derived from rice husk, organic solid wastes and sewage sludge, J. Environ. Manage. 133 (2014) 309–314. https://doi.org/10.1016/j.jenvman.2013.12.007.

[24] T. Goldammer, Organic Crop Rotation, First Edit, Apex Publishers, 2017.

[25] IFOAM - Organics International, Organic Agriculture & Pesticides, 2020.

[26] IFOAM - Organics International, Comparisons Between Organic and Conventional Agriculture, 2020. (n.d.). https://www.ifoam.bio/news/comparisons-between-organic-and-conventional (accessed September 1, 2021).

[27] IFOAM - Organics International, Definition of Organic Agriculture | IFOAM, (2020). https://www.ifoam.bio/why-organic/organic-landmarks/definition-organic (accessed August 29, 2021).

Received: 21/06/2021, Accepted: 19/07/2021, Published:01/01/2022